A / From a code position to a goal readout

Extract selected attention-head contributions, then use the same model in a separate verbalization run.

(a) SOURCE COMPUTATION

TASK INSTRUCTIONS

Keep palindromes that start with a vowel.

CODE PREFIX

```
ys = [s for s in xs
      if s == s[::-1]]

zs =
```

t: activation at the final "=" token

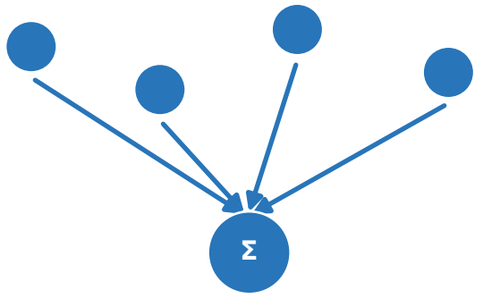
Completed: palindrome filter

Remaining: vowel-initial filter

(b) EXTRACT & AGGREGATE

Selected heads H, at token t

across source layers



$$g_t = \sum_{(\ell,h)\in H} W_O^{\ell,h}\, a_t^{\ell,h}$$

Project → sum → normalize

(c) PROBE GENERATION

FIXED TEXT PROMPT

"What are you doing at this line …?"

Source code is not in the probe prompt.

$\widehat{g}_t$

⊕ ▶ probe residual stream

⊕ ▶ probe residual stream

⊕ ▶ probe residual stream

"Keep strings starting with a vowel."

$$r_{\ell,s} \leftarrow r_{\ell,s} + \alpha \|r_{\ell,s}\| \widehat{g}_t$$

Same source vector at each selected probe layer and generation step.

DIFF readout settings: α = 0.1; layers 35–45; steering throughout decoding.

SCHEMATIC EXAMPLES · Verbalizations are illustrative, not recorded outputs. | Additive norm-relative GoalScope.

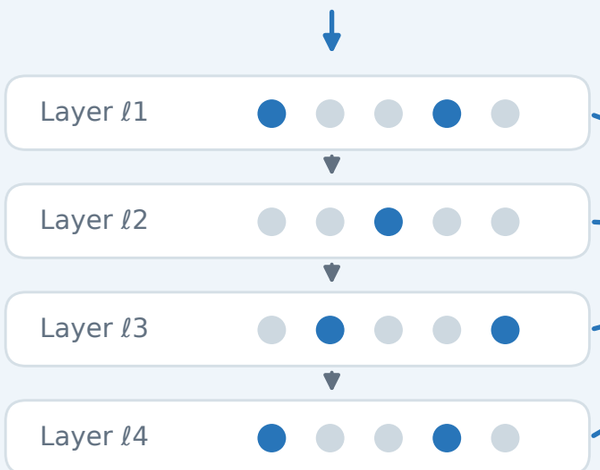
B / Two model runs, connected by one activation vector

Source head outputs are pooled across layers; the pooled direction is added at multiple probe layers.

(a) SOURCE RUN

Keep palindromes starting with a vowel.

```
ys = [s for s in xs
      if s == s[::-1]]
zs =
```



Read selected heads at one token t

Extraction layers determined by H

PROJECT & SUM

$$g_t = \sum_{(\ell,h)\in H} W_O^{\ell,h} a_t^{\ell,h}$$

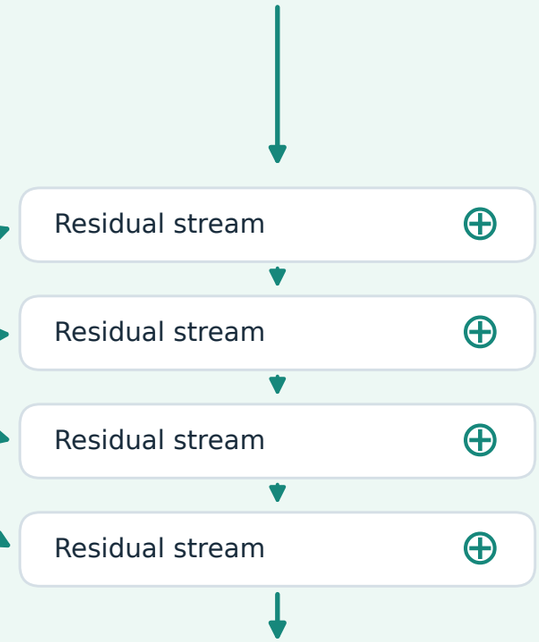
Normalize to $\hat{g}_t$

Arrows carry projected head outputs, not text or source residual states.

(b) PROBE RUN · SAME MODEL

“What are you doing at this line …?”

Fixed prompt; no source code text



“Keep vowel-initial strings.”

Depth ↓ · Layers shown schematically

$$r_{\ell,s} \leftarrow r_{\ell,s} + \alpha \|r_{\ell,s}\| \hat{g}_t$$

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C / Follow the goal as the computation progresses

Each marked token supplies a separate activation snapshot to the same fixed GoalScope probe.

SHARED INSTRUCTIONS Keep strings that are palindromes and start with a vowel; return the filtered list.

(a) SOURCE PREFIX AT EACH SNAPSHOT	(b) GOALSCOPE	(c) ILLUSTRATIVE READOUT
1 `ys =` **t1**	Head outputs → g_t Inject into fixed probe	**“Filter palindromes.”**
2 `ys = [s for s in xs if s == s[::-1]]` `zs =` **t2**	Head outputs → g_t Inject into fixed probe	**“Keep vowel-initial strings.”**
3 `ys = [s for s in xs if s == s[::-1]]` `zs = [s for s in ys if s[:1].lower() in "aeiou"]` `return` **t3**	Head outputs → g_t Inject into fixed probe	**“Return the filtered list.”**

Extraction tokens: t_1 = final “=” · t_2 = final “=” · t_3 = final token of “return”.

Each row is an independent probe generation; source snapshots are not accumulated inside the probe.

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D / A shared readout for clean and DIFF-patched computations

Hold the destination prompt fixed. Change the source computation, then extract its selected-head signal.

SHARED DESTINATION

Tasks: T_1 = palindrome; T_2 = vowel-initial

```
ys = [s for s in xs if s == s[::-1]]
zs =      ← extract at the final "=" token t
```

DONOR SETUPS

Q donor: same {T_1, T_2}; completed {T_2}.

KV donor: {T_1, T_3}; completed {T_1}.

T_3 = more than one distinct character.

(a) COMPUTATION INSPECTED	(b) EXTRACT	(c) FIXED PROBE	(d) READOUT
Clean destination Original destination forward	Selected head sum g_t^{clean}	Same prompt + model $r \leftarrow r + \alpha \lVert r \rVert \hat{g}_t$	**Keep vowel-initial strings.**
Query-patched destination Transplant donor Q at selected heads	Selected head sum $g_t^{Q\ \text{patch}}$	Same prompt + model $r \leftarrow r + \alpha \lVert r \rVert \hat{g}_t$	**Filter palindromes.**
KV-patched destination Transplant donor task-span K,V	Selected head sum $g_t^{KV\ \text{patch}}$	Same prompt + model $r \leftarrow r + \alpha \lVert r \rVert \hat{g}_t$	**Keep strings with >1 distinct character.**

The DIFF transplant occurs in the computation being inspected. GoalScope is a separate activation-to-text intervention.

Rows show predicted readouts for the illustrative setup, not measured transfer results or behavioral continuations.

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